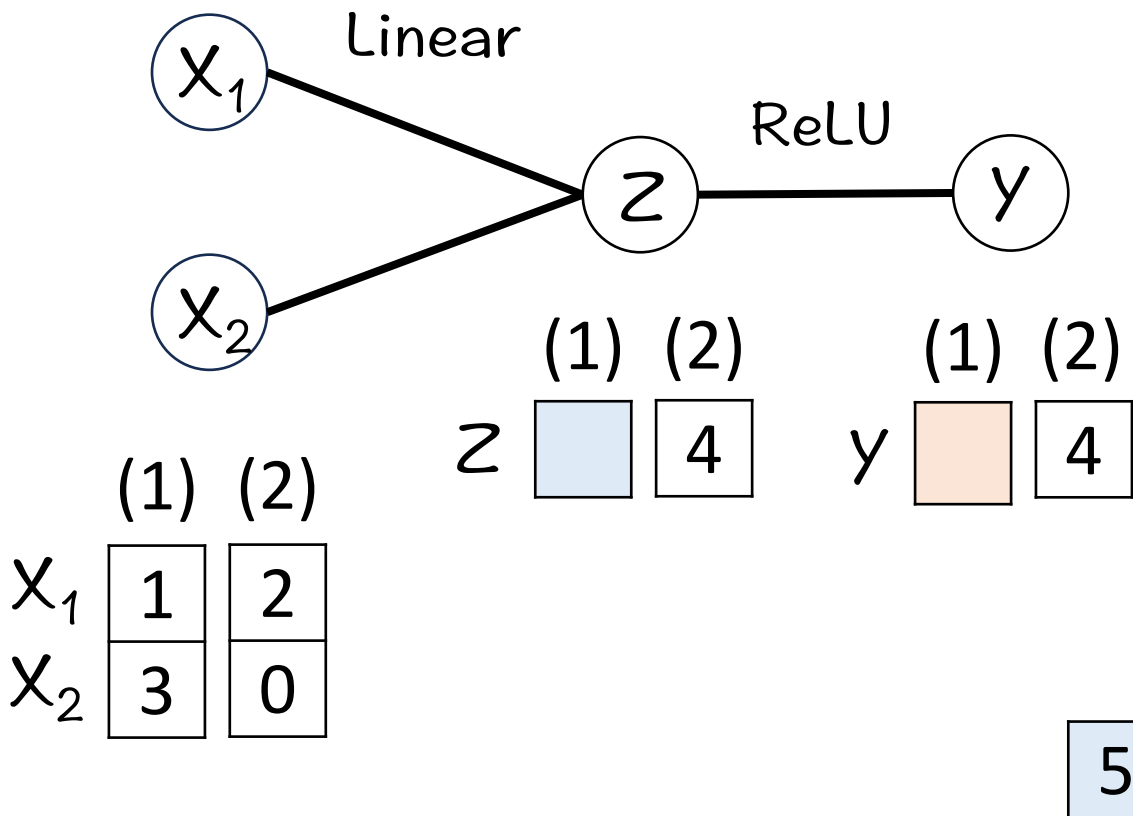
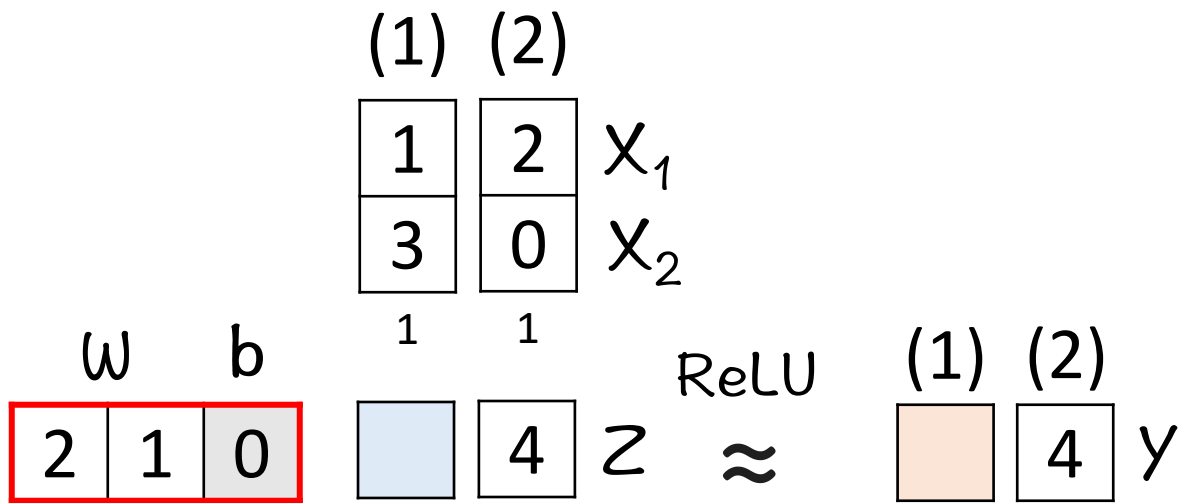


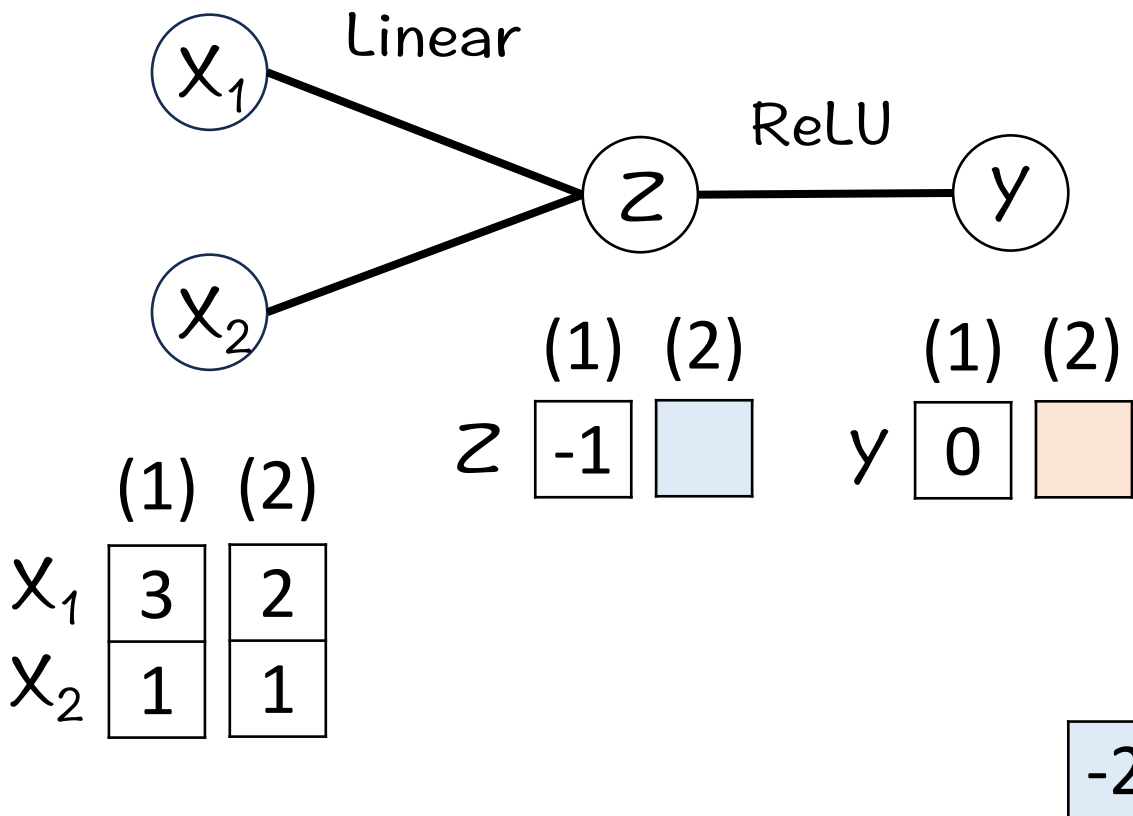
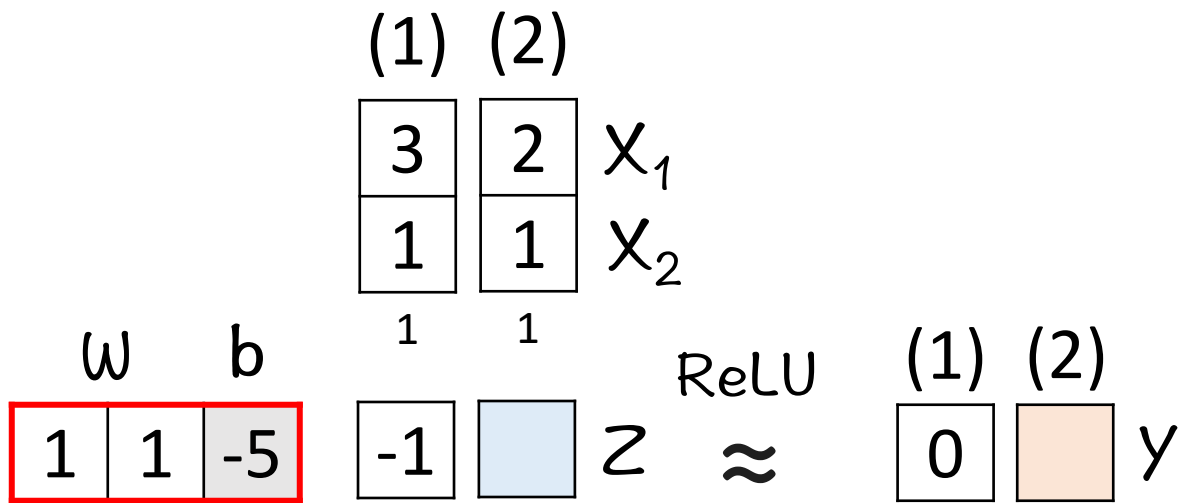
Batch

Exercise 1



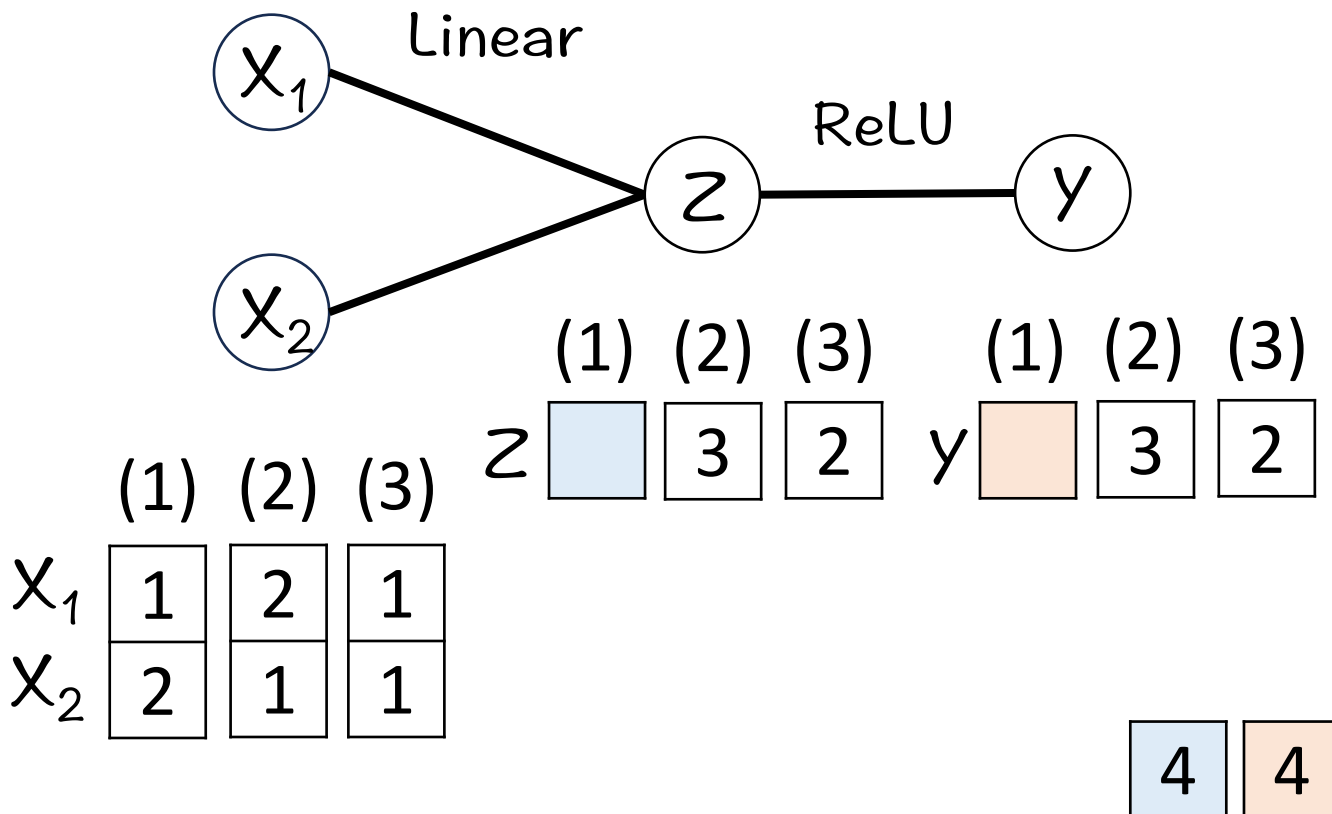
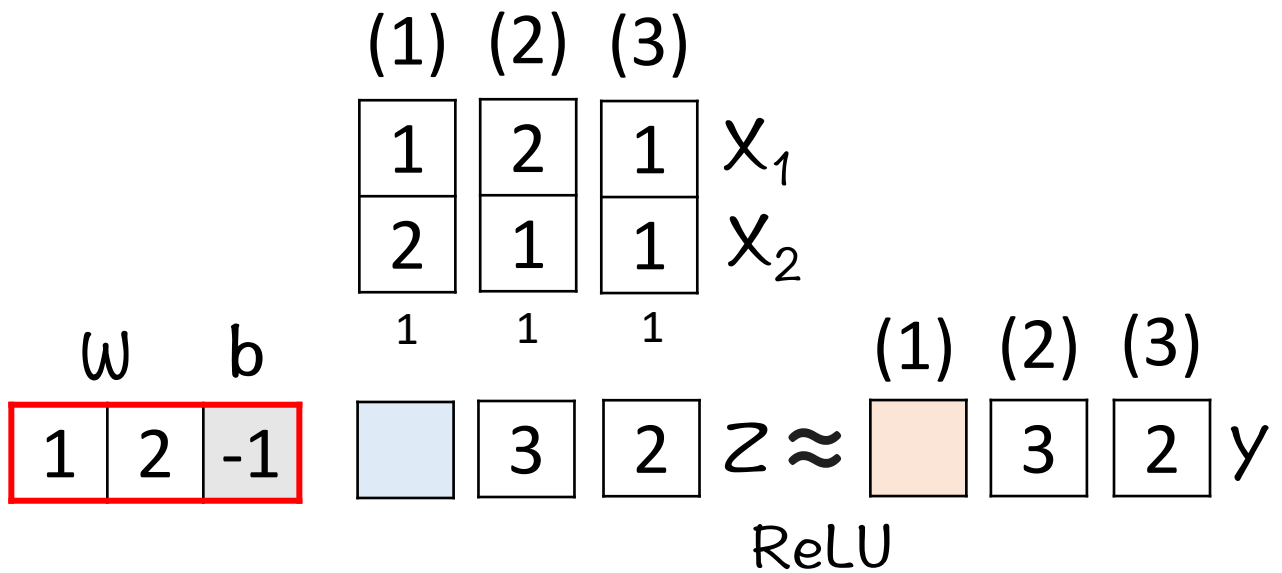
Batch

Exercise 2



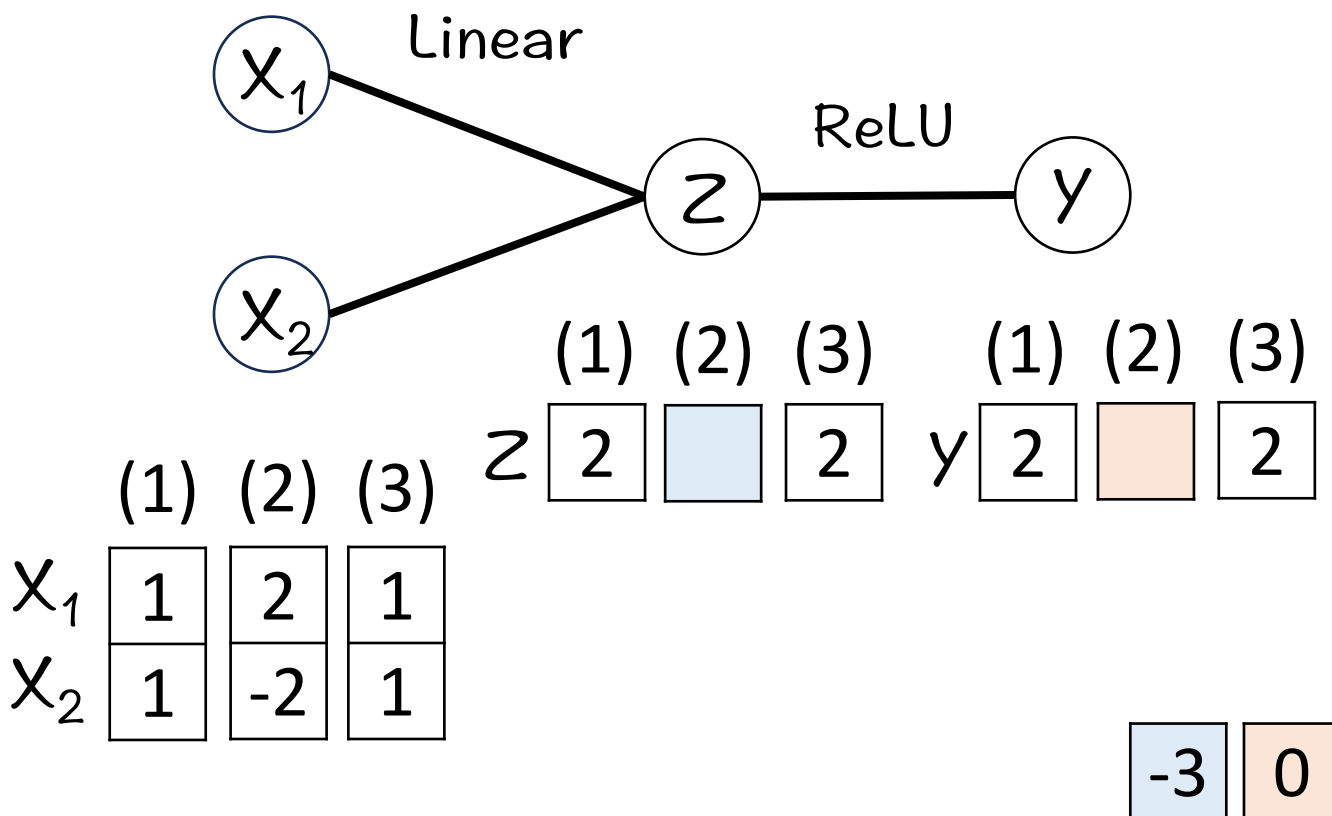
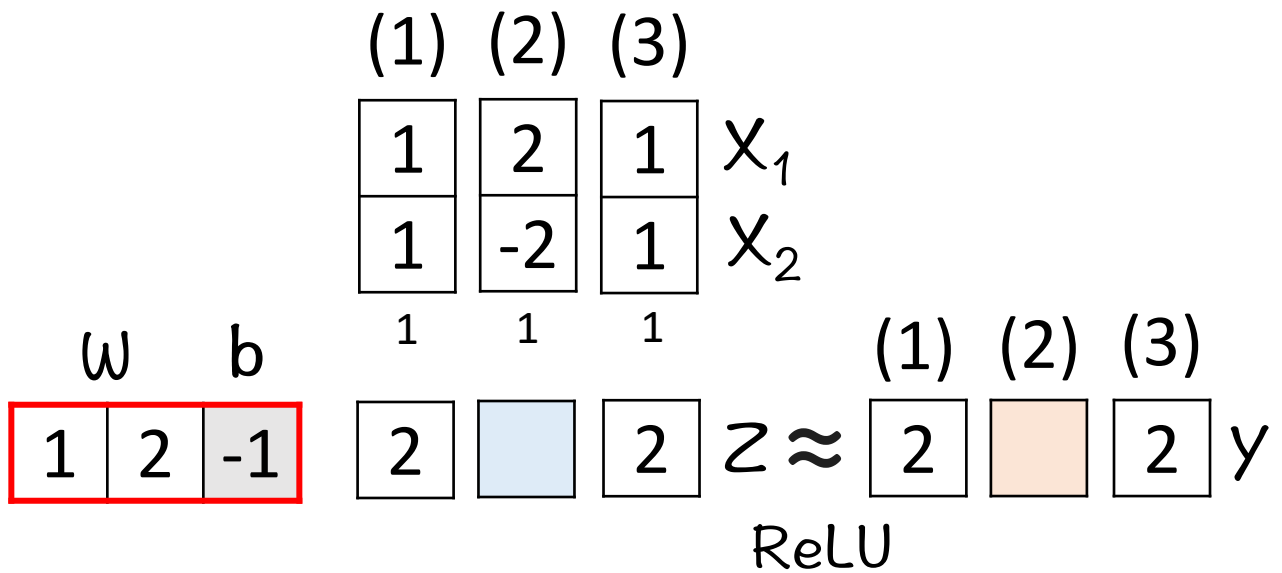
Batch

Exercise 3



Batch

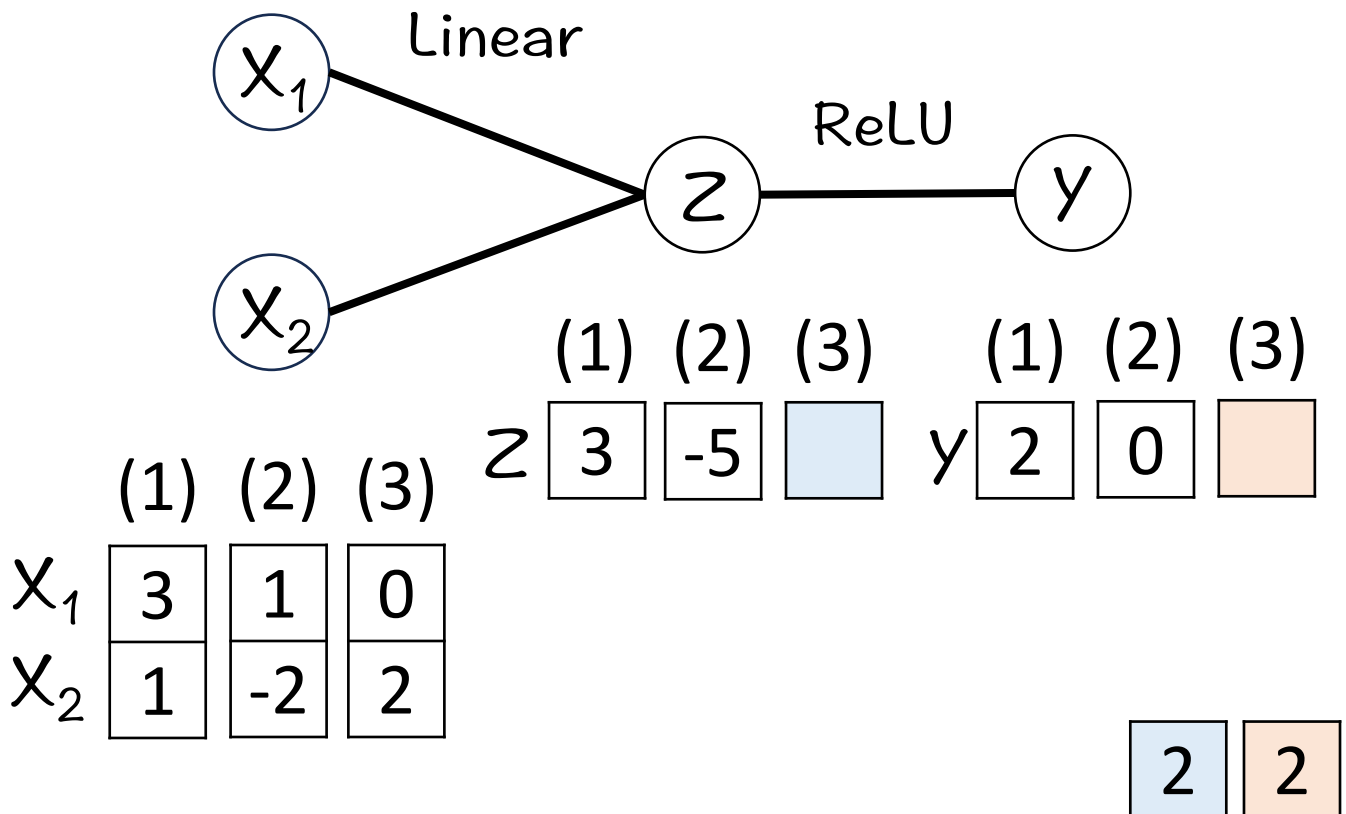
Exercise 4



Batch

Exercise 5

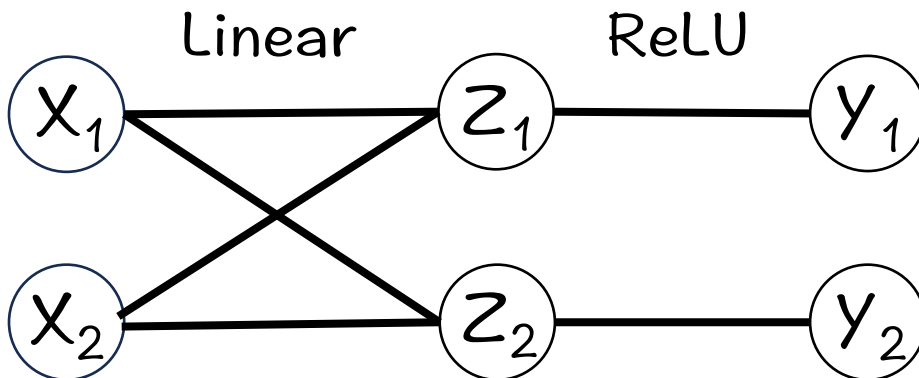
		(1)	(2)	(3)	
		3	1	0	x_1
		1	-2	2	x_2
w	b	1	1	1	
1	2	-2			
		3	-5		$z \approx$
		2	0		
					y
					ReLU



Batch

Exercise 6 (1) (2)

			(1)		(2)
			1	1	
			3	-4	x_1
			1	2	x_2
w	b				
1	0	2	5		z_1
0	1	-1	0	1	z_2
					ReLU $\approx$
			5		y_1
			0	1	y_2

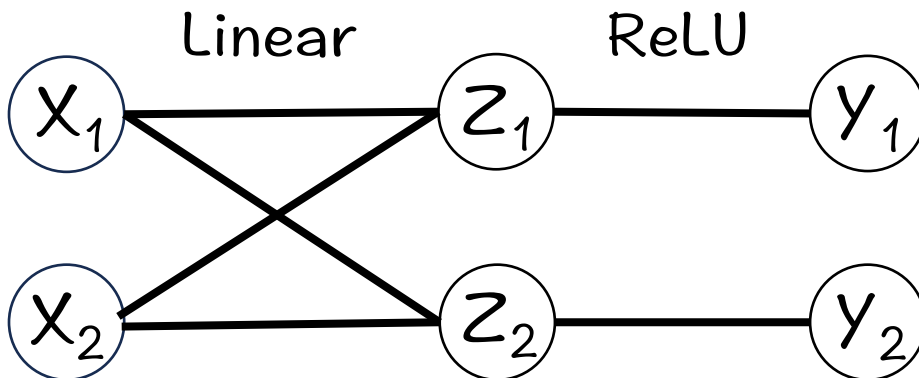


		(1)	(2)			(1)	(2)			(1)	(2)
		1	2			1	2			1	2
x_1		3	-4	z_1		5		y_1		5	
x_2		1	2	z_2		0	1	y_2		0	1
										-2	
										0	

Batch

Exercise 7 (1) (2)

			(1)	(2)		(1)	(2)	
			3	2	x_1	4	3	y_1
			1	3	x_2	0		y_2
			1	1				
w	b		4	3	z_1	ReLU		
1	0	1	0		z_2	$\approx$		
0	1	-2						

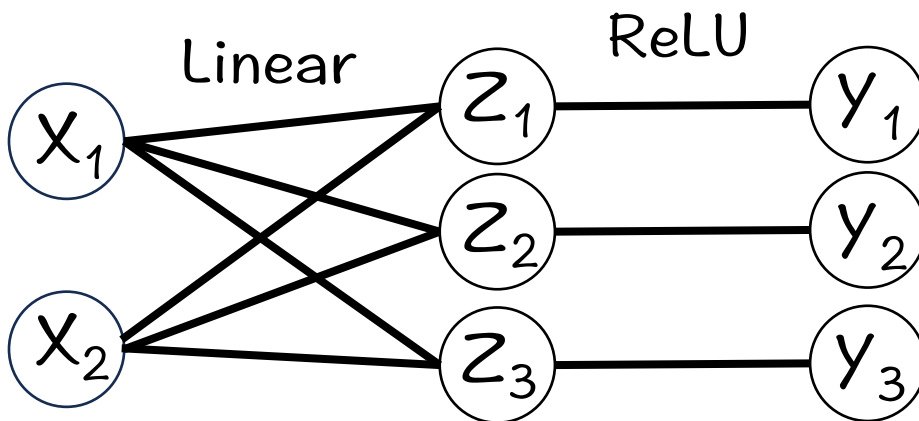


	(1)	(2)		(1)	(2)		(1)	(2)	
x_1	3	2	z_1	4	3	y_1	4	3	1
x_2	1	3	z_2	0		y_2	0		1

Batch

Exercise 8 (1) (2)

			(1)	(2)		(1)	(2)	
			3	-3	x_1			
			2	2	x_2			
w	b		1	1				
1	0	1	4		z_1	ReLU	4	
1	-1	0	1	-5	z_2	$\approx$	1	0
0	1	-1	1	1	z_3		1	1
								y_1
								y_2
								y_3

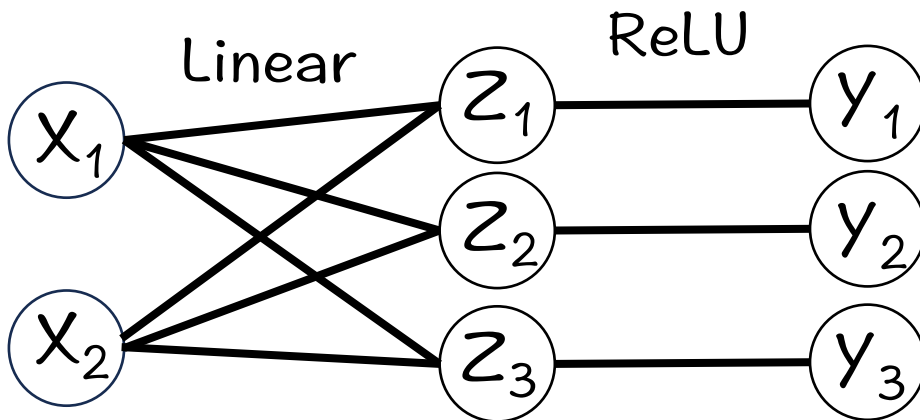


		(1)	(2)			(1)	(2)			(1)	(2)	
x_1		3	-3	z_1		4		y_1		4		
x_2		2	2	z_2		1	-5	y_2		1	0	-2
				z_3		1	1	y_3		1	1	0

Batch

Exercise 9

			(1)	(2)	(3)	(4)	
			3	2	3	2	X_1
			1	3	1	-2	X_2
w	b		1	1	1	1	ReLU
1	1	0	4		4	0	$Z_1 \approx$
1	-1	0	2	-1	2	4	Z_2
0	1	1	2	4	2	-1	Z_3
			(1)	(2)	(3)	(4)	
			4		4	0	Y_1
			2	0	2	4	Y_2
			2	4	2	0	Y_3



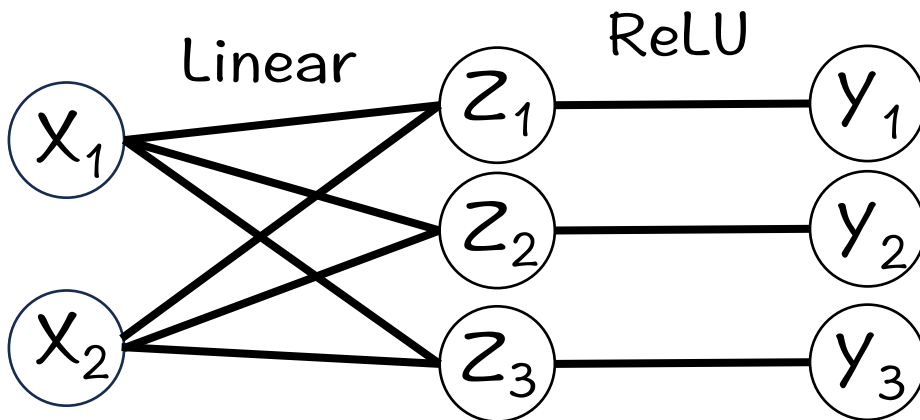
5

5

Batch

Exercise 10

			(1)	(2)	(3)	(4)	
			3	2	3	2	X_1
			1	1	2	-2	X_2
w	b		1	1	1	1	ReLU
1	1	0	4	3	5	0	$Z_1 \approx Y_1$
1	-1	0	2	1		4	$Z_2 \approx Y_2$
0	1	1	2	2	3	-1	$Z_3 \approx Y_3$



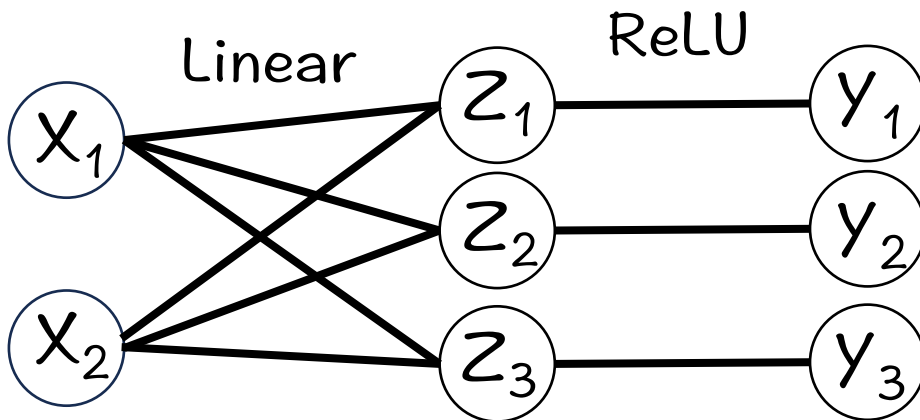
1

1

Batch

Exercise 11

			(1)	(2)	(3)	(4)	
			3	2	3	2	X_1
			1	2	1	-2	X_2
w	b		1	1	1	1	ReLU
1	1	0	4	4	4	0	$Z_1 \approx Y_1$
1	-1	0	2	0	2	4	$Z_2 \approx Y_2$
0	2	-1	1	3	1		$Z_3 \approx Y_3$



-5

0

Batch

Exercise 12

Batch Sum

		(1)	(2)	(3)	(4)
z_1	ReLU y_1	4	0	4	0
z_2	$\approx y_2$	2	0	2	6
z_3	y_3	2	3	0	-3

Σ (1)-(4)
8
2

Batch

Exercise 13

Batch Sum

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	Σ (1)-(10)
y_1	1	1	1	0	0	0	0	0	0	0	3
y_2	0	0	0	1	0	1	0	1	0	1	4
y_3	0	1	0	1	1	1	1	0	0	1	

6

Batch

Exercise 14

Batch Sum

Σ

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(1)-(10)
y_1	0	1	1	0	0	0	0	0	0	0	2
y_2	0	0	0	0	0	0	0	1	1	1	3
y_3	0	1	0	1	1		1	0	0	1	5

0

Batch

Exercise 15

Batch Mean

		(1)	(2)	(3)	(4)	μ (1)-(4)	
z_1	ReLU	y_1	4	0	4	0	2
z_2	$\approx$	y_2	2	2	2	6	
z_3		y_3	2	3	2	-3	1

3

Batch

Exercise 16

Batch Mean

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	μ (1)-(10)
y_1	1	1	1	0	0	0	0	1	0	0	0.4
y_2	0	1	0	1	0	1	0	1	0	1	0.5
y_3	1	1	0	1	1	1	1	0	0	1	

0.7

Batch

Exercise 17

Batch Mean

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	μ (1)-(10)
y_1	1	1	0	0	0	0	0	1	0	0	0.3
y_2	0	1	0	1	0	1	0	1	0	1	0.5
y_3	1		0	1	1	1	1	1	0	1	0.8

1

Batch

Exercise 18

Batch Add

$$\begin{array}{c} Y_1 \\ Y_2 \\ Y_3 \end{array} \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 4 & 0 & 4 & 0 \\ \hline 2 & 2 & 2 & 6 \\ \hline 2 & 3 & 2 & -3 \\ \hline \end{array} \cdot + \begin{array}{|c|} \hline 2 \\ \hline 1 \\ \hline 2 \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 6 & 2 & 6 & 2 \\ \hline 3 & 3 & 3 & 7 \\ \hline 4 & 5 & 4 & \text{ } \\ \hline \end{array}$$

-1

Batch

Exercise 19

Batch Add

	(1)	(2)	(3)	(4)
y_1	4	0	4	0
y_2	2	3	1	2
y_3	2	3	2	3

$$.+ \begin{array}{|c|} \hline 2 \\ \hline \\ \hline 1 \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline 6 & 2 & 6 & 2 \\ \hline 4 & 5 & 3 & \\ \hline 3 & 4 & 3 & 4 \\ \hline \end{array}$$

2

4

Batch

Exercise 20

Batch Subtract

$$\begin{array}{c} Y_1 \\ Y_2 \\ Y_3 \end{array} \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 4 & 0 & 4 & 0 \\ \hline 2 & 2 & 2 & 6 \\ \hline 2 & 3 & 2 & -3 \\ \hline \end{array} \cdot \begin{array}{|c|} \hline 2 \\ \hline 1 \\ \hline 1 \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 2 & -2 & 2 & -2 \\ \hline 1 & 1 & 1 & \text{ } \\ \hline 1 & 2 & 1 & -4 \\ \hline \end{array}$$

Batch

Exercise 21

Batch Subtract

	(1)	(2)	(3)	(4)
y_1	4	0	4	0
y_2	2	2	2	4
y_3	2	3	2	-3

 $\cdot$

2
1

 $=$

(1)	(2)	(3)	(4)
2	-2	2	-2
1	1	1	3
3	4	3	

-1

-2

Batch

Exercise 22

Batch Multiply

$$\begin{array}{c} Y_1 \\ Y_2 \\ Y_3 \end{array} \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 4 & 0 & 4 & 0 \\ \hline 2 & 1 & 2 & -1 \\ \hline 2 & 4 & 3 & -2 \\ \hline \end{array} \cdot * \begin{array}{|c|} \hline 2 \\ \hline 3 \\ \hline -1 \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 8 & 0 & 8 & 0 \\ \hline 6 & 3 & 6 & -3 \\ \hline -2 & & -3 & 2 \\ \hline \end{array}$$

-4

Batch

Exercise 23

Batch Multiply

$$\begin{array}{c} Y_1 \\ Y_2 \\ Y_3 \end{array} \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 4 & 0 & 4 & 0 \\ \hline 2 & 1 & 2 & -1 \\ \hline 2 & 4 & 3 & -2 \\ \hline \end{array} \cdot * \begin{array}{|c|} \hline 2 \\ \hline \text{ } \\ \hline 1 \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 8 & 0 & 8 & 0 \\ \hline 4 & 2 & 4 & \text{ } \\ \hline 2 & 4 & 3 & -2 \\ \hline \end{array}$$

2

-2

Batch

Exercise 24

Batch Center at Zero

$$\begin{array}{c} y_1 \\ y_2 \\ y_3 \end{array} \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline -2 & 0 & -4 & -2 \\ \hline 3 & 6 & 3 & 0 \\ \hline -1 & 1 & 1 & -1 \\ \hline \end{array} \cdot - \begin{array}{|c|} \hline \mu \\ (1)-(4) \\ \hline -2 \\ \hline \text{ } \\ \hline 0 \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 0 & 2 & -2 & 0 \\ \hline 0 & 3 & \text{ } & -3 \\ \hline -1 & 1 & 1 & -1 \\ \hline \end{array}$$

3

0

Batch

Exercise 25

Batch Center at Zero

$$\begin{array}{c} y_1 \\ y_2 \\ y_3 \end{array} \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 4 & 0 & 4 & 0 \\ \hline 4 & 2 & 1 & 1 \\ \hline 1 & 4 & -1 & 0 \\ \hline \end{array} \cdot - \begin{array}{|c|} \hline \mu \\ \hline (1)-(4) \\ \hline 2 \\ \hline 2 \\ \hline \text{ } \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline (1) & (2) & (3) & (4) \\ \hline 2 & -2 & 2 & -2 \\ \hline 2 & 0 & -1 & -1 \\ \hline 0 & 3 & \text{ } & -1 \\ \hline \end{array}$$

1

-2